# Neea Rusch's Dissertation Defense

## Applied Implicit Computational Complexity

**Abstract** Implicit computational complexity (ICC) complements classic complexity theory by developing machine-independent characterizations of complexity classes. The idea is to introduce a restriction, at the level of a programming language, that guarantees every program satisfying the restriction belongs to a particular complexity class. There are several strong motivations for this approach, including that ICC systems drive better understanding of complexity classes, yield natural definitions and proofs, and produce techniques for automatable program analysis for free. However, despite the numerous advantages, ICC remains largely a theoretical novelty. This means the practical power, limitations, and advantages of ICC-based program analyses are not well-understood. The dissertation research "puts ICC theories to test" by investigating their capabilities outside the theoretical domain. A guiding intuition is that, if applied, implicit computational complexity could provide us new techniques for program analysis and verification.

| | |
|---|---|
| **Keywords** | programming languages, implicit computational complexity, formal methods |
| **Degree** | Doctor of Philosophy |
| **Date** | 📅 Friday, 15 August 2025 |
| **Time** | 🕙 10 AM – 1 PM Eastern Time |
| **Add to Calendar** | `neea.pl/defense.ics` |
| **Physical Venue** | Dr. Roscoe Williams Ballroom<br>Room 155, Jaguar Student Activities Center (JSAC)<br>Summerville Campus |
| **Virtual Venue** | `augusta-edu.zoom.us/j/97369163776` |

## Committee

- Clément Aubert, *major advisor*, Augusta University
- Martin Avanzini, Centre Inria d'Université Côte d'Azur, France
- Yuyan Bao, Augusta University
- Bogdan Chlebus, Augusta University
- Harley Eades III, Augusta University

## Presentation Preview

Refer to `neea.pl/dissertation` for all available documents. For a brief overview, see the *extended abstract* that summarizes the full-length dissertation. The dissertation is open source.

A copy of this announcement is available at `neea.pl/defense`.

# Candidate Introduction

Neea Rusch is a long-term member of the Augusta University community. She entered the university as an undergraduate in 2011 and graduated as The Outstanding Computer Science Student of the Class of 2014. Upon graduation, she accepted a position as a software engineer and simultaneously continued her education at the Georgia Institute of Technology. At Georgia Tech, she was a member of the inaugural cohorts of the Online Master of Science in Computer Science (OMSCS) program. Since her graduation from the OMSCS, more than 10,000 students worldwide have followed in the same footsteps. Therefore, it was only natural that when the *second* opportunity to pioneer a computer science graduate program surfaced at Augusta University, Neea signed up immediately. At conclusion of the doctoral program, she will become a double Jag and a legacy graduate. Neea has also been an instructor at Augusta University. In that role she has taught over 250 students the principles of programming. Several of her former students have already earned graduate degrees and even joined the computer science doctoral program along with their former instructor.

During her doctoral studies, Neea has been actively involved in the programming languages research community. Among her many merits are serving in the program committees of 23 international conferences, winning 3x the title of a distinguished artifact reviewer, and becoming the editor of the Journal of Open Source Software. At Augusta University, Neea is the co-founder of ΔΛΔ, a student organization dedicated to promoting interest in programming languages research. Since January 2023, the organization has been hosting the weekly programming languages reading group and other related activities. With over 50 hosted events so far, ΔΛΔ is the oldest and most active graduate-level computer science student organization at Augusta University.

Finally, Neea has been a vigorous representative of the computer science doctoral program and its students. In this capacity, Neea has broken trails and acted as the voice of her fellow students. For example, Neea has served in multiple graduate student committees (TGS-GSC and GSGA) for a total of six years. In 2024, Neea participated in the 3 Minute Thesis (3MT) research communication competition as the first contestant with a computing-themed pitch. This effort then led to the opportunity to present the pitch in Atlanta, at the University System of Georgia headquarters. But Neea's activities are not just limited to Georgia. As a frequent traveler, Neea has assumed the role of an international student ambassador. From Norway to New Zealand, and San Francisco to Singapore, Neea has represented Augusta University at various research events in 24 cities, in 12 countries, and across 4 continents. Not surprisingly, her post-graduation plans are 4,500 miles away.

#### Education

- 8/2011–5/2014 Bachelor of Science, Computer Science, Augusta University
- 8/2014–5/2016 Master of Science, Computer Science, Georgia Institute of Technology
- 8/2021– * Doctor of Philosophy, Computer & Cyber Sciences, Augusta University

#### Select Academic Achievements

- 3x Distinguished Artifact Reviewer award at ECOOP 2025, TACAS 2025, and OOPSLA 2024
- 3 Minute Thesis (3MT) 2024 finalist, Augusta University
- Acquired over $30K in academic and research funding, from multiple awards
- Co-founder, former Vice President, and current member of ΔΛΔ
- Excellence in Research in Computer & Cyber Sciences 2022, Augusta University
- Grace Hopper Scholarship 2015, Georgia Institute of Technology
- The Outstanding Computer Science Student, Class of 2014, Augusta University

# On the Day of the Defense

## Preliminary Schedule

| Time | Description |
| --- | --- |
| 9:00 AM | Venue doors open |
| 10:00 AM | Public presentation and question/answer session |
| 11:00 AM | *Halftime*<br>Committee and candidate adjourn to rounds of questioning<br>Audience is entertained by activities in the ballroom |
| approx. 12:30 PM | Closed committee meeting to determine examination outcome |
| approx. 1 PM | Announcement of results |
| 1 PM | Social hour |
| 3 PM | Closing |

The planned and reserved halftime activities include popcorn machines, a cornhole board, cotton candy, snow cones, and games.

## Organizing Committee

The organization of the defense is generously supported by the following student volunteers.

- Gabriele Cecilia, *Audio Engineering Specialist*
- Peter Hanukaev, *Chief Guest Relations Officer*
- Mahady Hassan, *Conference Concierge*
- Mark Holcomb, *Conference Concierge*
- Vignesh Sivakumar, *Audio Engineering Specialist*
- Jason Weeks, *Popcorn Maestro*

## Attending the Virtual Presentation

The broadcast will be delivered live over Zoom. The conferencing software is available for download at `zoom.us/download`. For optimal experience, online participants should install the software in advance.

## Attending the Physical Event

**Important:** aim to arrive around 9:45 AM.

Dr. Roscoe Williams Ballroom is located in the first floor of the Jaguar Student Activities Center, on the Summerville campus. The campus street address is 2500 Walton Way, Augusta, Georgia 30904.

## Driving Directions and Parking

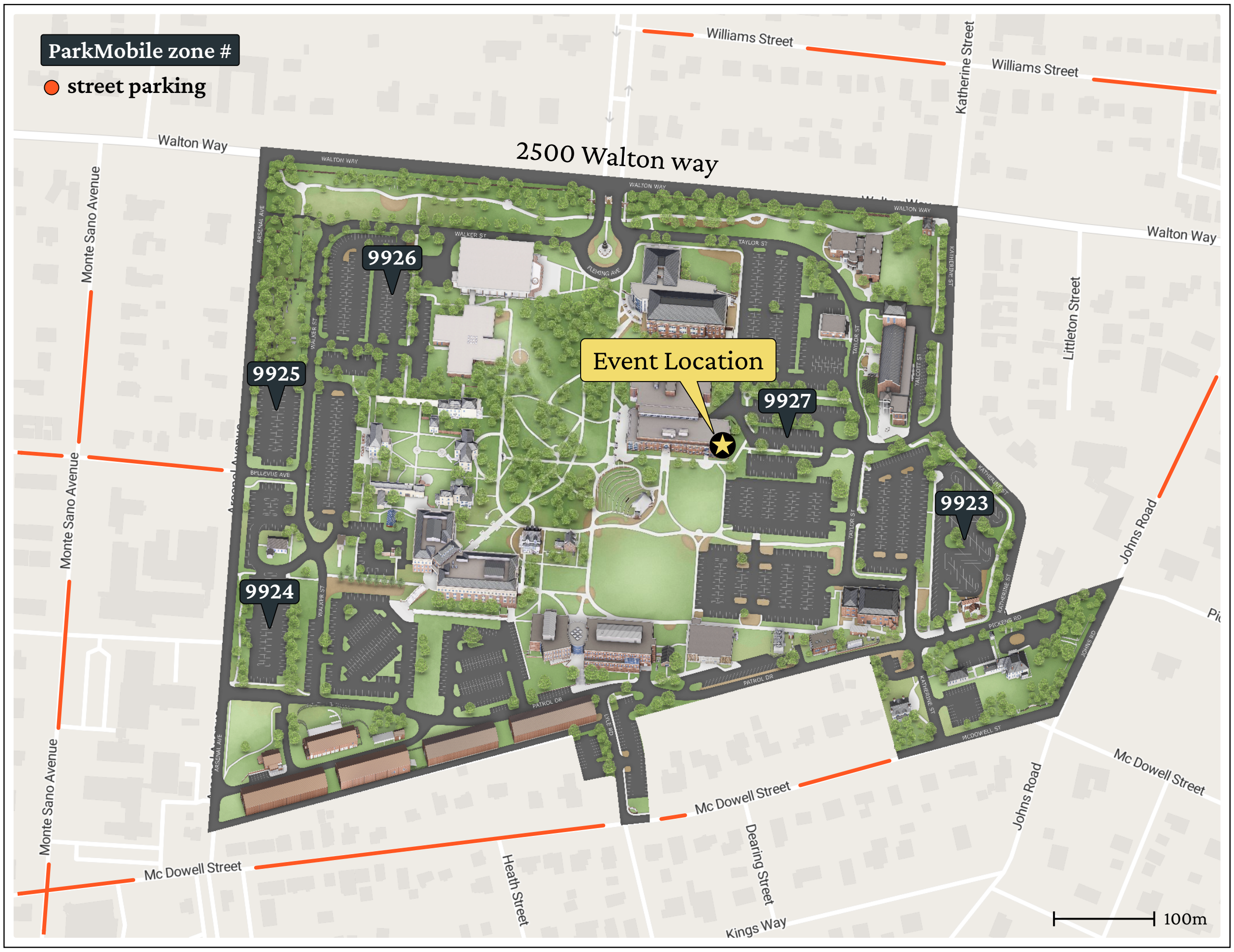
For directions to the Jaguar Student Activities Center, see:

google.com/maps/dir/Current+Location/33.47633862058983,-82.02202358771989

For typing convenience, tinyurl.com/5emv7cb3 is a shorter version of the same URL.

**Street parking** is free and available on many streets around the Summerville campus. Suggestions of where to find free parking are annotated in the map.

**Parking on campus** is by permit only, Monday – Friday, from 8 AM to 5 PM. Without a permit, make sure to pay for parking, or you can get a ticket. Use the ParkMobile app or go to parkmobile.io for payment. The ParkMobile zone codes for the available lots are annotated in the map.




## JagExpress Shuttle Service

A shuttle service is available for transportation between campuses. The shuttle arrives to the Summerville Fountain by the main entrance. For real-time shuttle locations, use the PassioGO app at passiogo.com.